

CVC Shared Tools

Almost-Zero-Cost Maintenance & Free Documentation for Climate Workflows

Prepared by an AI agent CVC

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Do you use these or this?

Do you use these?



OpenAI



Gemini



Claude



Copilot



Meta AI



DeepSeek

VS

Or do you use this?



BSC Institutional Repository

Rev. v1.0

cvc-shared-tools

gitlab.earth.bsc.es/cvc/cvc-shared-tools.git

The Irony: Many of us interact with AI tools daily. Yet our shared departmental code remains scattered across personal /gdfs/ scratch/ directories where it quickly decays.

Machine learning is revolutionizing weather forecasting – the next step is a change in how we work

Peter Dueben^{a,*1}, Peter Bauer^{b,2}, Oliver Fuhrer^{c,d,3}, Nikolay Koldunov^{e,4} and Jørn Kristiansen^{f,5}

^aEuropean Centre for Medium-Range Weather Forecasts, Robert-Schumann-Platz 3, 53175 Bonn, Germany

^bMax-Planck-Institute for Meteorology, Bundesstrasse 53, Hamburg, 20146, Germany

^cFederal Office of Meteorology and Climatology MeteoSwiss, Operation Center 1, Airport, Zürich, 8058, Switzerland

^dInstitute for Atmospheric and Climate Science, ETH, Sälimstrasse 101, Zürich, 8092, Switzerland

^eAlfred-Wegener-Institute for Polar and Marine Research, Am Handelshafen 12, Bremerhaven, 27570, Germany

^fNorwegian Meteorological Institute, Postboks 43 Blindern, Oslo, 0313, Norway

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ABSTRACT

Following the success of machine learning in producing weather predictions with competitive skill compared to complex traditional systems, this article shifts attention from forecast output to the working practices that make prediction systems possible. We argue that machine learning and recent digital technologies will reshape the forecasting value chain: how models are coded and developed, how observations and Earth-system data are exploited, how data and computing are managed, how systems are verified, and how information is created, evaluated and turned into services. We discuss six non-exhaustive areas in which agentic software engineering, open and compressed data, shared verification workflows, interactive computing and generative methods may make modelling, evaluation and service creation faster, more interactive and more widely accessible. These changes will require weather and climate centres to adapt their infrastructures, data stewardship, trust and quality-assurance frameworks, skills and service delivery while maintaining scientific understanding, operational reliability, human expertise and their public-service role.

Dueben (ECMWF), Bauer (MPI-M), Fuhrer (MeteoSwiss), Koldunov (AWI), Kristiansen (Met Norway)

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1. Paradigm Shift: Ways of Working

SHIFT

- ML skill is proven; the next step is **reshaping working practices**.
- Re-engineering how climate models and workflows are coded, verified, and maintained.

2. Agentic Software Engineering

AGENT

- Shifting burden from individual postdocs to **clear contracts, automated tests, and agentic workflows**.
- Requires robust data stewardship and trust frameworks.

AI climate change: Adaptation or Mitigation

Where Scripts Live Today

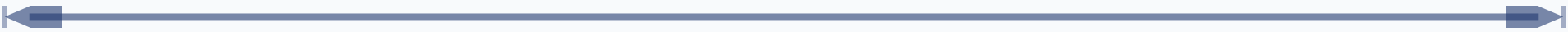
SCRATCH

- **Personal scratch spaces:** Scripts live in `/gdfs/scratch/bsc32/...`, unshared and unversioned.
- **Copy-paste variants:** Ten slightly different versions of the same interpolation script.
- **Silent breakage:** Hardcoded paths and modules break quietly when HPC systems update.

Why Manual Docs Fail

DOCS

- **Wikis detach from code:** Documentation written in wikis rots within weeks.
- **High maintenance friction:** Scientists don't have time to manually update documentation after parameter tweaks.
- **Unknown health:** Nobody knows if an old script still runs until someone tries it.



The Core Opportunity

We rarely lose scientific workflows because the physics was wrong. We lose them because documentation was detached from code. We need documentation that rebuilds itself automatically in Git.

Everything in Git

REPO

- workflows/**/metadata.yaml: Structured metadata validated by schema.
- workflows/**/scripts/: Actual Bash, Python, or CDO scripts.
- docs/decisions/: Scientific & infrastructure Decision Records (ADRs).
- WORKFLOWS.md: Automatically generated workflow catalog.
- schemas/: Machine-executable contracts.

Core Rule: No Hidden State

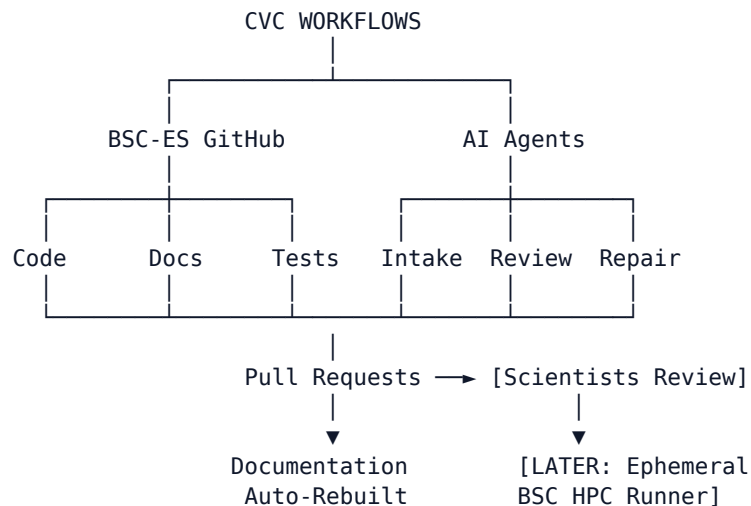
If it is not in Git, it does not exist for the team or for assistants.
Git provides complete transparency and version history.

metadata.yaml Contract

Rev. CONTRACT

```
schema_version: "1.0"
workflow:
  name: "en4-to-orca"
  tier: "production"
environment:
  platform: "mn5"
  modules:
    - "netcdf/4.7"
    - "cdo"
    - "sosie"
validation:
  documentation: "reviewed"
  software: "passed"
  scientific: "required"
```

The Problem is Institutional Memory



How We Preserve Memory: By connecting shared code and AI intake through standard Pull Requests, scientists stay in control while documentation rebuilds automatically on merge.

A “Smart Intern” in the Cloud

HOW IT WORKS

- **Beyond rigid scripts:** Traditional CI only runs fixed commands. Agentic workflows (gh-aw) give an AI a goal, tools, and repo context.
- **Can read and reason:** It reads messy user scripts, checks schema rules, spots deprecated HPC modules, and drafts docs.
- **Triggered automatically:** Runs on schedule, on issue creation, or on new PRs via standard GitHub Actions.

Strict Safety & Zero Cost

SAFETY

- **Can only suggest, never merge:** All agent output is restricted to Draft PRs or Issue comments for scientists to review.
- **Zero HPC credentials:** No SSH access, no keys, zero execution on MareNostrum 5 or cluster nodes.
- **Free maintenance:** Runs entirely on GitHub free runner tiers and API allocations (Gemini / Claude).

In short: Think of gh-aw as an eager research assistant who reads through messy scripts, fills out the paperwork, and leaves a tidy draft on your desk for approval.

Deterministic Catalog Generator

INDEXER

- **Zero manual editing:** A lightweight Python script (`update_workflows_index.py`) scans all workflow metadata.
- **Instantaneous & reliable:** Validates against JSON Schema and rebuilds `WORKFLOWS.md` in **under 1 second**.
- **Zero server cost:** Published automatically via GitHub Pages / ReadTheDocs on every merge to `main`.
- **Always in sync:** When a scientist updates `metadata.yaml`, the public documentation updates immediately.

What Scientists & Engineers Get

BENEFIT

- **Instant Discovery:** Search workflows by tool (`cd0`, `NEMO`, `SOSIE`, `xarray`).
- **HPC Compatibility:** Filter immediately by platform (`mn5`, `workstation`).
- **Trust Transparency:** Clear badges for documentation, software checks, and scientific review.

The Win: Scientists never maintain separate wiki pages. Documentation stays permanently linked to code and rebuilds itself automatically on every merge.

Three Distinct Trust Tiers

Rev. TRUST

- **1. Software Health (Deterministic):** ShellCheck warnings = 0, Python syntax passes, schema valid.
- **2. Documentation Completeness:** Clear inputs/outputs, platform requirements, and contact authors.
- **3. Scientific Truth (Human-Reviewed):** Grid interpolation, constants, and physics confirmed by domain scientists.

Strict Safety Boundaries (Class C)

Rev. SAFETY

- **Zero HPC Execution:** Agents never hold SSH keys, credentials, or submit SLURM jobs to MareNostrum 5.
- **Never Alter Physics:** AI cannot autonomously alter numerical schemes, grid definitions, or physical constants.
- **Sole Gatekeeper:** All agent actions are gated behind human review via standard Pull Requests.

Core Rule: We never trust a single green CI check to declare science valid. Software health, documentation, and scientific verification are kept strictly decoupled.

Pragmatic Engineering (YAGNI & DRY)

DESIGN

- **No heavy web portal:** Git is the durable storage and single source of truth.
- **No custom CLI runtime:** Standard `git`, `bash`, and `python`. No bespoke daemons to install or maintain.
- **Tool Neutrality:** Developers can use any assistant locally (Antigravity, Claude Code, Copilot) or GitHub cloud agents.

Seeding the Commons: The First 10

ROADMAP

- **Focus on high-reuse workflows:** Onboard 10 essential preprocessing pipelines (ocean regridding, atmospheric masking, CDO chains).
- **Clean reference standards:** Each workflow provides tested scripts, `metadata.yaml`, and sample inputs.
- **Measured by reuse:** The goal is not lines of code, but developer hours saved and scripts rescued from decay.

Design Motto: Keep it flat, keep it in Git, and start small. If a workflow isn't saving real developer time on MN5, it doesn't belong in the repo.

Production Prototype: en4-to-orca

WORKFLOW

- **Path:** workflows/en4-to-orca/
- **Function:** SOSIE 3D interpolation of ocean temperature & salinity.
- **Target Platform:** MareNostrum 5 (netcdf/4.7, cdo, sosie).
- **Verified Clean:** 0 ShellCheck warnings, strict bash flags (set -euo pipefail), documented modules.

0

SHELLCHECK WARNINGS

< 1s

CATALOG BUILD TIME

Repository Machine Contracts

Rev. CONTRACTS

- schemas/workflow.schema.json: Strict JSON schema validating workflow metadata.
- schemas/extraction.schema.json: Schema for agent knowledge extraction.
- docs/decisions/: Architecture & scientific Decision Records (ADRs).
- AGENTS.md: Machine-readable contract defining trust tiers and safety boundaries for all agents.

Ready Today: The repository contract, schemas, catalog generator, and reference workflow are already implemented and running in this repository.

1. Health Audit (repo-health)

Rev. AUDIT

- **Trigger:** Scheduled weekly (Monday) via GitHub Actions (gh-aw).
- **Agent Action:** Runs Gemini/Claude in a sandbox, reads all workflows, checks schemas & module deprecations.
- **Safe Output:** Opens a GitHub Issue with a health matrix and actionable checklist.
- **Live Proof:** Issue #10 was generated live by Claude Sonnet 4.6 in **65 seconds** at zero cost!

2. Ingestion (inspect-workflow)

Rev. INGEST

- **Trigger:** Scientist opens an issue requesting workflow onboarding.
- **Agent Action:** Parses script paths, extracts facts (T0), flags unknowns, and drafts `metadata.yaml`.
- **Safe Output:** Posts a structured review comment on the issue.
- **Draft PR:** Proposes a clean Draft Pull Request for scientist review — human merges to `main`.

Safe Automation: AI agents do the tedious reading, schema checking, and drafting via Issues and Draft PRs. Scientists retain 100% control over merging to `main`.



The project succeeds when the repository becomes more useful than the sum of the scripts inside it.



cvc-shared-tools

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Questions & Discussion